

Literature Review – FocusBot

AI-Powered Interactive Desk Companion for Productivity and Well-Being

Literature Review 1

Title: A Distributed Multi-Modal Sensing Approach for Human Activity Recognition in Real-Time Human-Robot Collaboration

Authors: Valerio Belcamino, Nhat Dinh Minh Le, Quan Khanh Luu, Alessandro Carfi, Van Anh Ho, Fulvio Mastrogiovanni

Publication: IEEE Robotics and Automation Letters, 2025

DOI / Reference Link: <https://doi.org/10.1109/LRA.2025.3592072>

Research Objectives

- Develop a human activity recognition (HAR) system for real-time human-robot collaboration.
- Combine complementary sensing modalities to capture human hand activities during interaction with a robot.
- Evaluate the activity-recognition approach in offline, real-time, and realistic human-robot collaboration settings.

Research Gaps

- Single sensing modalities may not provide sufficient information for reliable activity recognition during physical human-robot interaction.
- Real-time collaboration requires robots to recognize human actions while interaction is occurring.
- More validation is needed for multimodal sensing in realistic collaborative scenarios.

Research Questions

- How effectively can multiple sensing modalities be combined for human activity recognition?
- Can the proposed system recognize activities in real time?
- How well does multimodal sensing perform in realistic human-robot collaboration?

Method / Approach

The study combines a modular data glove equipped with inertial measurement units (IMUs) and a vision-based tactile sensor. The system is evaluated through segmented offline classification, real-time classification, and a realistic human-robot collaboration scenario.

Main Findings

The experiments reported high recognition accuracy across the tested tasks, supporting the usefulness of multimodal sensing for collaborative robots.

Relevance to FocusBot

For FocusBot, the study supports the use of multiple sensory inputs to recognize user activities and make the robot responsive to the user's current context.

Literature Review 2

Title: Natural Multimodal Fusion-Based Human–Robot Interaction: Application With Voice and Deictic Posture via Large Language Model

Authors: Yuzhi Lai, Shenghai Yuan, Youssef Nassar, Mingyu Fan, Atmaraaj Gopal, Arihiro Yorita, Naoyuki Kubota, Matthias Rätzsch

Publication: IEEE Robotics & Automation Magazine, 2025/2026 publication record

DOI / Reference Link: <https://doi.org/10.1109/MRA.2025.3543957>

Research Objectives

- Develop a natural human-robot interaction framework using more than one communication modality.
- Combine voice commands with deictic/pointing posture information.
- Use a large language model (LLM) to translate multimodal human intent into robot actions.

Research Gaps

- Voice-only interaction can be difficult when commands are complex.
- Gesture-only interaction may not provide enough semantic information to identify the user's complete intention.
- Existing HRI systems need more natural ways to combine visual and verbal context.

Research Questions

- How can voice and pointing information be fused for natural HRI?
- Can an LLM use temporally aligned visual and verbal information to infer human intent?
- Does multimodal interaction improve task accuracy and robustness?

Method / Approach

Visual information is processed using object detection and depth information to identify relevant objects. Voice is converted to text, and the selected visual information is temporally aligned with the command. An LLM then generates robot action sequences while control syntax constraints are used to reduce unsafe or invalid outputs.

Main Findings

The real-world evaluation with a Universal Robots UR3e demonstrated improved HRI performance in terms of accuracy and robustness, according to the paper's reported results.

Relevance to FocusBot

FocusBot can apply the same principle by combining what the camera sees with what the user says, allowing Gemini/LLM reasoning to produce context-aware responses.

Literature Review 3

Title: AURA: Advancing Human-Robot Interaction Through Unified Voice and Gesture Control

Authors: AURA project authors

Publication: 2025 8th International Conference on Electronics, Materials Engineering & Nano-Technology (IEMENTech)

DOI / Reference Link: <https://ieeexplore.ieee.org/document/10959702/>

Research Objectives

- Develop an AI assistant capable of interacting through voice and air gestures.
- Integrate computer vision and natural language processing into a physical robot.
- Provide hands-free and voice-free interaction options.

Research Gaps

- Many traditional AI assistants depend primarily on a single input modality such as voice.
- Voice interaction is not always convenient or possible in every environment.
- Gesture recognition can provide an additional interaction channel, but it needs to be integrated with voice-based interaction.

Research Questions

- How can voice and gesture recognition be unified in one robot assistant?
- Can the robot dynamically interpret both vocal and gesture-based commands?
- Can multimodal control improve flexibility and accessibility?

Method / Approach

The system uses a Raspberry Pi, camera, speaker, and microphone. Computer vision is used for gesture recognition and natural language processing is used for voice interaction.

Main Findings

The project demonstrates unified voice and gesture control and shows how a compact hardware platform can support multimodal human-robot interaction.

Relevance to FocusBot

This is directly useful for FocusBot's hardware and interaction design because FocusBot also needs a camera, microphone, speaker, AI processing, and physical responses.

Literature Review 4

Title: Enabling Conversational Intelligence in Service Robots Through Multimodal Fusion and Resource-Aware Generative Models

Authors: T. Tharmilan, Henoshan Athimoolanathan, Garthigan Kumarasamy, A. G. Buddhika P. Jayasekara, V. Logeeshan, Chathura Rajakaruna Wanigasekara

Publication: 2025 IEEE World AI IoT Congress (AlIoT)

DOI / Reference Link: <https://ieeexplore.ieee.org/document/11105269/>

Research Objectives

- Develop a multimodal conversational framework for service robots.
- Fuse visual and speech information for context-aware communication.
- Use generative AI, including LLMs/VLMs, to support socially intelligent interaction.
- Consider resource constraints for near-real-time deployment.

Research Gaps

- Traditional service robots have limited conversational and contextual understanding.
- Generative AI models can require substantial computational resources.
- Robots need to combine speech, vision, emotion, and context rather than treating each input independently.

Research Questions

- How can vision and speech be fused for context-aware robot conversation?
- Can generative AI support adaptive and socially responsive HRI?
- How can multimodal generative models be deployed on resource-constrained robotic platforms?

Method / Approach

The work presents a modular multimodal architecture using vision-language models such as LLaVA together with speech-recognition and face-detection components. Quantized models and efficient deep-learning components are considered for constrained deployment.

Main Findings

The paper reports preliminary module-integration results and identifies the potential of multimodal generative AI for adaptive, explainable, and collaborative HRI. The authors describe the deployment as experimental/work in progress.

Relevance to FocusBot

FocusBot follows a very similar pipeline: camera and speech inputs are interpreted by an AI model, which then produces a context-aware response.

Literature Review 5

Title: MindEye-OmniAssist: A Gaze-Driven LLM-Enhanced Assistive Robot System for Implicit Intention Recognition and Task Execution

Authors: Zejia Zhang, Bo Yang, Xinxing Chen, Weizhuang Shi, Haoyuan Wang, Wei Luo, Jian Huang

Publication: 2025 IEEE International Conference on Cyborg and Bionic Systems (CBS)

DOI / Reference Link: <https://arxiv.org/abs/2503.13250>

Research Objectives

- Develop an assistive robot that can recognize implicit human intentions.
- Use gaze information together with vision and an LLM to infer user intent.
- Generate robot action sequences from inferred intentions rather than relying only on explicit commands.

Research Gaps

- Many assistive robots depend on explicit commands from users.
- Gaze-based systems have often been limited to simple or predefined assistance tasks.
- More flexible intention recognition is required for diverse, less-defined tasks.

Research Questions

- Can gaze information help a robot infer implicit user intention?
- Can an LLM expand intention recognition beyond predefined commands?
- Can inferred intentions be converted into useful robot action sequences?

Method / Approach

The system combines an open-vocabulary object detector, an intention-recognition network, gaze/eye-movement information, and an LLM. The system was tested in real-world assistive tasks.

Main Findings

The reported real-world experiments achieved 41 successful tasks out of 55 across various undefined assistive tasks.

Relevance to FocusBot

FocusBot also aims to infer the user's state from observed behavior rather than requiring manual input for every action. The paper supports the idea of implicit, context-aware interaction.

Literature Review 6

Title: Estimating User Engagement in Human Robot Interaction Using a Dynamic Bayesian Network

Authors: Xiaoxuan Hei, Heng Zhang, Adriana Tapus

Publication: 2025 IEEE International Conference on Robotics and Automation (ICRA)

DOI / Reference Link: <https://ieeexplore.ieee.org/document/11128383/>

Research Objectives

- Estimate user engagement during human-robot interaction.
- Combine multiple behavioral and physiological/visual features.
- Develop a model that can classify users as engaged or not engaged during interaction.

Research Gaps

- User engagement is difficult to estimate continuously from a single signal.
- Human engagement can be reflected through several behavioral and facial indicators.
- Robots need methods that can integrate multiple signals to estimate engagement dynamically.

Research Questions

- Can user engagement be estimated automatically during HRI?
- Which visual and physiological features are useful for engagement estimation?
- Can multimodal information improve engagement classification?

Method / Approach

A Dynamic Bayesian Network is used to combine head rotation, eye movements, facial expressions, and facial temperature measured using a thermal camera. Data came from an HRI experiment in which a robot guided participants.

Main Findings

The proposed model achieved an accuracy of 0.83 and an AUC of 0.82 for classifying participants as engaged or not engaged.

Relevance to FocusBot

FocusBot can use similar engagement concepts to distinguish focused study from distraction or disengagement.

Literature Review 7

Title: Temporal and Modality Awareness-Based Lightweight Residual Network With Attention Mechanism for Human Activity Recognition Using a Lower-Limb Exoskeleton Robot

Authors: Chang-Sik Son, Won-Seok Kang

Publication: IEEE Access, 2025

DOI / Reference Link: <https://scholar.dgist.ac.kr/handle/20.500.11750/58906>

Research Objectives

- Develop a lightweight neural network for human activity recognition.
- Recognize locomotion activities using multimodal sensing data.
- Capture temporal and modality-specific features while reducing computational complexity.

Research Gaps

- Many high-accuracy HAR models have high computational complexity.
- Resource-constrained robotic systems require efficient models.
- Different sensor modalities contain different temporal information that should be represented effectively.

Research Questions

- Can a lightweight network maintain strong activity-recognition performance?
- How can temporal and modality-specific features be extracted efficiently?
- Can model complexity be reduced for deployment on constrained robotic hardware?

Method / Approach

The proposed network uses an asymmetric convolutional architecture with depthwise and pointwise layers and an attention mechanism to capture temporal and modality-specific features efficiently.

Main Findings

The study reports a lightweight HAR model designed to reduce trainable parameters while retaining effective recognition of diverse locomotion activities across different terrains.

Relevance to FocusBot

FocusBot is intended to be a low-cost desktop robot, so computational efficiency is important. Lightweight activity recognition can make real-time deployment more practical.

Literature Review 8

Title: Human Activity Recognition by Using Enhanced Radar Point Cloud 2D Histograms and Doppler Feature Fusion

Authors: Guanghang Liao, Jieming Ma, Fei Luo

Publication: 2025 IEEE International Conference on Robotics and Automation (ICRA)

DOI / Reference Link: <https://ieeexplore.ieee.org/document/11128165/>

Research Objectives

- Develop a non-invasive human activity recognition framework using millimeter-wave radar.
- Improve radar point-cloud representations using spatial information.
- Fuse point-cloud and Doppler features for more comprehensive activity recognition.

Research Gaps

- Conventional radar point-cloud histograms may not fully represent relative spatial changes.
- Doppler information can provide additional movement-related information.
- Better feature fusion is needed for robust radar-based HAR.

Research Questions

- Can enhanced point-cloud histograms improve human activity recognition?
- Can Doppler features provide complementary information?
- Can feature fusion improve recognition accuracy for robotics and HCI applications?

Method / Approach

The method integrates fixed radar sensing boundaries into 2D point-cloud histograms and concatenates range-Doppler and angle-Doppler histograms. A hybrid network is used to address overfitting and optimize the number of stacked layers.

Main Findings

The reported accuracy was 99.72% on the mmWaveRadarWalking dataset and 98.67% on the CI4R-Human-Activity-Recognition dataset.

Relevance to FocusBot

FocusBot currently relies on camera-based perception, but radar could be a future privacy-preserving sensing option for recognizing activity without continuously recording images.

Literature Review 9

Title: Help or Hindrance: Understanding the Impact of Robot Communication in Action Teams

Authors: Tauhid Tanjim, Jonathan St. George, Kevin Ching, Hee Rin Lee, Angelique Taylor

Publication: 2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)

DOI / Reference Link: <https://doi.org/10.1109/RO-MAN63969.2025.11217909>

Research Objectives

- Investigate how robot communication affects human workload and perceptions.
- Study verbal and non-verbal robot communication in a time-sensitive task environment.
- Examine how multimodal cues can support humans in remembering tasks and searching for objects.

Research Gaps

- Robots need to communicate information without creating unnecessary workload.
- Less is known about effective multimodal robot communication in time-sensitive team environments.
- The timing and modality of robot feedback can influence how useful the robot is to users.

Research Questions

- How do verbal robot cues affect human workload and perceptions?
- How do visual cues affect task reminders?
- Can multimodal robot feedback improve perceived usefulness and ease of use?

Method / Approach

The study used an experimental medical-training scenario in which a robotic crash cart provided verbal and non-verbal cues for iterative tasks and supply-search activities.

Main Findings

The study reports that verbal cues for object-search tasks and visual cues for task reminders reduced team workload and increased perceived ease of use and usefulness compared with a robot providing no feedback.

Relevance to FocusBot

This strongly supports FocusBot's use of context-aware reminders instead of fixed notifications. FocusBot can choose whether to use voice, display, lights, or movement depending on the user's situation.

Literature Review 10

Title: Development of a Humanoid Robot Prototype for Multimodal Human-Robot Interaction

Authors: Thang Tran Viet, Thanh Nguyen Canh, Huy Uong Gia, Phuc Dinh Van, Son Tran Duc, Ngoc Minh Do, Xiem HoangVan

Publication: 2025 RIVF International Conference on Computing and Communication Technologies (RIVF)

DOI / Reference Link: <https://ieeexplore.ieee.org/document/11364526/>

Research Objectives

- Develop a physical humanoid robot as a flexible testbed for multimodal HRI.
- Integrate gesture recognition, object detection, speech recognition, and LLM-based semantic parsing.
- Validate the robot platform through physical positioning and interaction experiments.

Research Gaps

- Integrated physical platforms are needed to combine several AI modules into one HRI system.
- Perception, speech, semantic reasoning, and physical interaction must work together for natural HRI.
- Accessible robotic platforms can help researchers prototype multimodal interaction systems.

Research Questions

- How can multiple AI perception and interaction modules be integrated into one physical robot?
- Can gesture and object recognition work together with voice interaction?
- Can an LLM support semantic interpretation of voice commands for robot actions?

Method / Approach

The platform uses a 12-DOF dual-arm mechanism, a 2-DOF head with an LCD display, onboard Jetson processing, MediaPipe Pose with an LSTM classifier for gestures, YOLO for object detection and 3D localization, and speech recognition with LLM-based semantic parsing.

Main Findings

The reported experiments achieved over 90% task accuracy, including 96% gesture recognition and 92% speech recognition. Average manipulation error was approximately 1.83 cm.

Relevance to FocusBot

This paper provides a useful reference for FocusBot's integrated architecture: camera perception, speech, LLM reasoning, display/expression, and physical robotic behavior.

Overall Research Gap for FocusBot

The reviewed studies demonstrate strong progress in human activity recognition, multimodal human-robot interaction, engagement estimation, LLM-based interaction, and robot communication. However, these capabilities are generally investigated as separate components or in domains such as industrial collaboration, healthcare, assistive robotics, and general HRI. A focused integration of computer vision, activity recognition, LLM-based contextual reasoning, voice interaction, and physical robotic feedback for everyday student productivity remains an opportunity for further research.

- Integrate real-time user activity recognition with an LLM-based decision layer.
- Distinguish productive activities from distraction and idle states.
- Provide context-aware interventions instead of only fixed reminders.
- Combine voice, display, lighting, and physical movement as adaptive feedback channels.
- Track study, break, distraction, and idle periods in a low-cost desktop robot.
- Adapt interventions to the user's activity and preferences.

Reference Links

1. A Distributed Multi-Modal Sensing Approach for Human Activity Recognition in Real-Time Human-Robot Collaboration — <https://doi.org/10.1109/LRA.2025.3592072>
2. Natural Multimodal Fusion-Based Human–Robot Interaction: Application With Voice and Deictic Posture via Large Language Model — <https://doi.org/10.1109/MRA.2025.3543957>
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